

CHE 312

HV 12

90/100

Aly Villalobos

1) A → B

$$R_{AB} = -k_B$$

A → C

$$R_{AC} = -k_C$$

A → F

$$R_{AF} = -k_F C_A$$

$$B.C. I = C_A = C_0 \text{ at } y=0 \quad \checkmark$$

$$D_{AB} \frac{d^2 C_A}{dy^2} = k_B + k_C + k_F C_A = 0 \quad \checkmark$$

$$B.C. II = -D_{AB} \left. \frac{dC_A}{dy} \right|_{y=L} = 0 \quad \checkmark$$

$$\frac{d^2 C_A}{dy^2} - \frac{k_B + k_C}{D_{AB}} C_A = \frac{k_B + k_C}{D_{AB}}$$

$$C_A(y) = A e^{y \sqrt{\frac{k_F}{D_{AB}}}} + B e^{-y \sqrt{\frac{k_F}{D_{AB}}}} - \frac{k_B + k_C}{D_{AB}}$$

Use B.C. I

$$C_A(0) = A e^0 + B e^0 - \frac{k_B + k_C}{D_{AB}} = C_0$$

$$C_A(0) = A + B - \frac{k_B + k_C}{D_{AB}} \quad \checkmark$$

$$A = C_0 - B + \frac{k_B + k_C}{D_{AB}} \quad \checkmark$$

Use B.C. 2

$$-D_{AB} \left. \frac{dA}{dy} \right|_{y=L} = 0$$

$$-D_{AB} \left[A \frac{\sqrt{\frac{k_F}{D_{AB}}}}{L} e^{-\frac{\sqrt{\frac{k_F}{D_{AB}}} y}{L}} - B \frac{\sqrt{\frac{k_F}{D_{AB}}}}{L} e^{-\frac{\sqrt{\frac{k_F}{D_{AB}}} y}{L}} + 0 \right] = 0$$

$$-A D_{AB} \frac{\sqrt{\frac{k_F}{D_{AB}}}}{L} e^{-\frac{\sqrt{\frac{k_F}{D_{AB}}} y}{L}} + B D_{AB} \frac{\sqrt{\frac{k_F}{D_{AB}}}}{L} e^{-\frac{\sqrt{\frac{k_F}{D_{AB}}} y}{L}} = 0$$

$$A D_{AB} \frac{\sqrt{\frac{k_F}{D_{AB}}}}{L} e^{-\frac{\sqrt{\frac{k_F}{D_{AB}}} y}{L}} = B D_{AB} \frac{\sqrt{\frac{k_F}{D_{AB}}}}{L} e^{-\frac{\sqrt{\frac{k_F}{D_{AB}}} y}{L}}$$

$$B = A e$$

$$A + A e = C_0 + \frac{k_B + k_c}{D_{AB}}$$

$$A \left[1 + e \right] = C_0 + \frac{k_B + k_c}{D_{AB}}$$

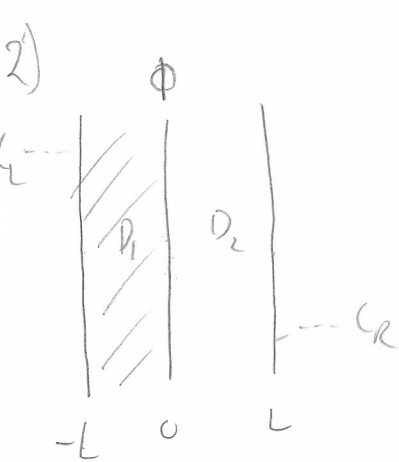
$$A = \frac{C_0 + \frac{k_B + k_c}{D_{AB}}}{\left[1 + e \right]}$$

$$B = \frac{C_0 + \frac{k_B + k_c}{D_{AB}}}{\left[1 + e \right]} e$$

$$G(y) = \frac{C_0 + \frac{k_B + k_C}{D_{AB}}}{1 + e^{\frac{2L\sqrt{\frac{k_F}{D_{AB}}}}} \left(e^{y\sqrt{\frac{k_F}{D_{AB}}}} \right) - \frac{C_0 + \frac{k_B - k_C}{D_{AB}}}{1 + e^{\frac{2L\sqrt{\frac{k_F}{D_{AB}}}}} \left(e^{2L\sqrt{\frac{k_F}{D_{AB}}}} \right) - y\sqrt{\frac{k_F}{D_{AB}}}} - \frac{k_B + k_C}{D_{AB}}$$



$N_A'' = ?$



B.C. 1 : $C_A(-L) = C_L$

B.C. 2 : $C_A(L) = C_R$

B.C. 3 : $\Phi_1 = \frac{C_A(0)}{C_{A2}(0)}$

$\Phi_1 = \frac{B}{F}$ $F = \frac{B}{\Phi_1}$

B.C. 4 : $-D_1 \left. \frac{dC_A}{dx} \right|_{x=0^-} = -D_2 \left. \frac{dC_{A2}}{dx} \right|_{x=0^+}$

$-D_1 \frac{d^2 C_{A1}}{dx^2} = 0$

$-D_2 \frac{d^2 C_{A2}}{dx^2} = -K_A = 0$

$-D_1 \frac{d^2 C_{A1}}{dx^2} = 0$

$-D_2 \frac{d^2 C_{A2}}{dx^2} = K_A$

$\int \frac{d^2 C_A}{dx^2} = \int 0$

$\frac{-D_2 \frac{d^2 C_{A2}}{dx^2}}{-D_2} = \frac{K_A}{D_2}$

$\int \frac{d^2 C_{A2}}{dx^2} = \int \frac{-K_A}{D_2}$

$C_{A1}(x) = Ax + B$

B.C. 1

$C_{A1}(-L) = -AL + B$

$C_{A2}(x) = \frac{-K_A x^2}{D_2} + Ex + F$

$C_L = -AL + B$

B.C. 2 : $C_A(L) = \frac{-K_A}{D_2} L^2 + EL + F$

$C_R = \frac{-K_A L^2}{D_2} + \frac{D_1 B}{D_2} + F$

$-D_1 B = -D_2 \left(\frac{-2K_A(0)}{D_2} + E \right)$ B.C. 4

$C_R = \frac{-K_A L^2}{D_2} + \frac{D_1 B}{D_2} + \frac{B}{\Phi_1}$

$+D_1 B = +D_2 E$

$C_R + \frac{K_A L^2}{D_2} = B \left(\frac{D_1}{D_2} + \frac{1}{\Phi_1} \right)$

$D_1 B = D_2 E$

$B = \frac{D_2 E}{D_1}$ $E = \frac{D_1}{D_2} B$

$B = \frac{C_R + \frac{K_A L^2}{D_2}}{\left(\frac{D_1}{D_2} + \frac{1}{\Phi_1} \right)}$

$E = \frac{D_1}{D_2} \left[\frac{C_R + \frac{K_A L^2}{D_2}}{\frac{D_1}{D_2} + \frac{1}{\Phi_1}} \right]$

$$F = \frac{\left(C_R + \frac{K_A L^2}{D_2} \right)}{\frac{D_1}{D_2} + \frac{1}{\phi_1}}$$

$$A = \frac{C_L - B}{-L}$$

$$A = \frac{C_L - \left(\frac{C_R + \frac{K_A L^2}{D_2}}{\frac{D_1}{D_2} + \frac{1}{\phi_1}} \right)}{-L}$$

$$C_{A1}(X) = \underbrace{C_L - \left(\frac{C_R + \frac{K_A L^2}{D_2}}{\frac{D_1}{D_2} + \frac{1}{\phi_1}} \right)}_{-L} X + \frac{C_R + \frac{K_A L^2}{D_2}}{\frac{D_1}{D_2} + \frac{1}{\phi_1}} \quad \checkmark$$

$$C_{A2}(X) = \frac{-K_A}{D_2} X^2 + \frac{D_1}{D_2} \left[\frac{C_R + \frac{K_A L^2}{D_2}}{\frac{D_1}{D_2} + \frac{1}{\phi_1}} \right] X + \left(\frac{C_R + \frac{K_A L^2}{D_2}}{\frac{D_1}{D_2} + \frac{1}{\phi_1}} \right) \quad \checkmark$$

$N_A'' = ?$